

Post-recombined terminal coordinate, fixed-six-row conormal Gram envelopes, sharp balanced bridge, and the direct-low response boundary

TECT verification-first repository

claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

first issued 2026-07-31 · this version issued 2026-07-31 · v1.0

1. Purpose and scope

R-129 isolated the legal aggregate reverse and the shifted-Douglas acceptance criterion, but left the owner-complete production response, forward constants, balanced coefficient, low block, and common terminal open. R-130 advances five parts of that frontier.

1. After the complete current has been recombined, the R-082 four-row coordinate Ξ admits an exact common-terminal Doob and once-owned trace recode.
2. The R-123 action owner has a canonical finite-cylinder response: form the physical bilinear form first and pull it back to predictable source space second. The executable proves this order on finite matrices; it does not construct a cutoff-uniform A1 response.
3. The fixed six-row A1 Gram has explicit pointwise first- and second-variation envelopes, with every factor of two fixed.
4. The deterministic balanced bridge

$$\int |z|^2 |\nabla z|^2 \leq \|\Delta z\|_2 \|z\|_6^3 \tag{1.1}$$

has sharp constant one.

5. A centered-heat direct-low packet admits a finite lower bound after one global Young inequality, while its relation to the historical R-079 low owners remains unproved.

The scope is fixed finite cutoff, fixed density floor $e = 10^{-12} > 0$, one fixed R-093/R-104 temporally faithful chart, a fixed conditional law, a fixed filtration, and an owner-complete post-recombined terminal packet. R-130 proves neither a bounded bare- L^2 production response uniform in cutoff, floor, chart, control, filtration, visit, shell, or refinement depth, nor complete C_{mix} , C_{far} , or c_{bal} data. It does not close the historical R-079 low owners, a strict augmented gap, the production common-terminal matching-energy theorem, the absolute anchor, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, the interacting measure, Sector A, or any T5–T7 claim.

2. Owner and coordinate firewall

The argument uses the following incidence rules.

Object	Ownership rule
$\mathcal{P}_{\text{comp}}$	Current-square/trace packet, once after exhaustive recombination.
Future variance	Once through $q_{\text{comp}} = q_{\text{CN}} - q_{\text{fut}}/2$; it is not a favorable reserve for the smaller action owner.
R-063 forest	Alternate coordinate for the primitive trace, never an additional payment.
Heat	Internal to $P_{\Sigma}B = \mathbb{E}_r[C^T C]$; C is not averaged before squaring.
Six rows and Ξ	Two representations of one energy, not two owners.
Cartan block	Aggregated once; the full-cross factor is 2680/729.
Sextic and source cost	Each used once. The source cost $9I/10$ stays outside a physical pullback with vertical kernel.
Forward and reverse	One adjoint/symmetrized owner, not two reserves.
$\Delta C_b, \Delta \Gamma_r$	Distinct synthesis and Wick-trace data.
Direct low	Not addable to R-079 conditional or complete low without an equivalence theorem.

No root, visit, shell, or refinement-depth count may enter a production constant. The older local 40/729 checksum does not cancel the live 2680/729 full-cross Cartan owner.

3. The post-recombined Ξ terminal

For $\Psi = (u_1, u_2, \chi)$ and $\dot{\Psi} = (v_1, v_2, w)$ put

$$\begin{aligned}
r &= |u|^2, & \rho &= r + |\chi|^2, & d &= \rho + e, \\
\dot{r} &= 2 \operatorname{Re}(u^* v), & \dot{\rho} &= 2 \operatorname{Re}(u^* v + \bar{\chi} w), & h &= u_1 v_2 - u_2 v_1.
\end{aligned} \tag{3.1}$$

The exact R-082 coordinate is

$$\Xi(\Psi, \dot{\Psi}) = \begin{pmatrix} \sqrt{c_0} \dot{r} \\ \sqrt{c_1} (\dot{r} - \alpha \frac{r}{d} \dot{\rho}) \\ 2\sqrt{c_0 + c_1} \operatorname{Re} h \\ 2\sqrt{c_0 + c_1} \Im h \end{pmatrix}, \tag{3.2}$$

where

$$P = 4 + 10^{-12}, \quad \alpha = \frac{5}{9}, \quad c_0 = \frac{3}{250P}, \quad c_1 = \frac{243}{8000P}. \tag{3.3}$$

The first row is $\dot{r}$, not $\dot{\rho}$, and the second row contains r/d , not ρ/d . The Pauli completeness relations give

$$\sum_A J_A^2 = \dot{r}^2 + 4|h|^2, \quad \sum_A L_A^2 = \left(\dot{r} - \alpha \frac{r}{d} \dot{\rho} \right)^2 + 4|h|^2, \tag{3.4}$$

so (3.2) is a global norm recoding, including $u = 0$.

Let Ξ_*, Ξ_0 be one fixed terminal pair, let P_ℓ be the low conditional projection, and let d_j be the increments of one fixed filtration. Put

$$R_j = d_j(\Xi_* - \Xi_0).$$

Martingale orthogonality yields

$$\begin{aligned}
\frac{1}{2} \mathbb{E}(\|\Xi_*\|^2 - \|\Xi_0\|^2) &= \frac{1}{2} \mathbb{E}(\|P_\ell \Xi_*\|^2 - \|P_\ell \Xi_0\|^2) \\
&\quad + \sum_j \left\{ \mathbb{E} \langle d_j \Xi_0, R_j \rangle + \frac{1}{2} \mathbb{E} \|R_j\|^2 \right\}.
\end{aligned} \tag{3.5}$$

If the once-owned covariance partition is

$$\Gamma = \Gamma_\ell + \sum_j \Delta\Gamma_j, \quad (3.6)$$

trace linearity gives the exact square-minus-trace version of (3.5). The primary four-atom fixture and an independent eight-atom, three-coordinate, four-level fixture check reconstruction, pairwise orthogonality, total and coordinate-shell Pythagoras, relative recoding, and trace linearity.

Boundary 3.1: this is not a legacy-root transport theorem

The compression from the fixed six rows to Ξ depends on the state. Conditional expectation therefore need not commute with it. An arbitrary legacy collection of root blocks does not acquire one common terminal merely because its terminal norms agree. The independent tower fixture has

$$\phi_1 = (-1, -1, 1, 1), \quad \phi_2 = (0, 2, 4, 6),$$

but

$$\mathbb{E}[\phi_2 \mid \mathcal{F}_1] = (1, 1, 5, 5) \neq \phi_1. \quad (3.7)$$

The legal repair is to recombine the complete terminal first and rerun Doob directly in Ξ space.

Boundary 3.2: matching trace is not matching energy

Let $C(z)$ and c be $\mathcal{A}$ -measurable, put $B = C^T C$, and for a fresh derivative G define

$$\mu = \mathbb{E}[G \mid \mathcal{A}], \quad \Sigma_c = \text{Cov}(G \mid \mathcal{A}).$$

Then

$$\begin{aligned} \mathbb{E} \left[\frac{1}{2} |C(z)(G + c)|^2 - \frac{1}{2} \text{Tr}(B(z)\Gamma) \mid \mathcal{A} \right] &= \frac{1}{2} |C(z)(c + \mu)|^2 \\ &\quad + \frac{1}{2} \text{Tr}[B(z)(\Sigma_c - \Gamma)]. \end{aligned} \quad (3.8)$$

The nonnegative matching-energy specialization requires $\mu = 0$ and $\Sigma_c = \Gamma$. Later feedback can change either quantity, leaving a signed covariance defect.

The correct heat lift is

$$\Xi_\Sigma(z, y)(r') = C(z + r')y, \quad \|\Xi_\Sigma(z, y)\|^2 = y^T (P_\Sigma B)(z)y. \quad (3.9)$$

In general

$$P_\Sigma B - (P_\Sigma C)^T (P_\Sigma C) = \mathbb{E}[(C - P_\Sigma C)^T (C - P_\Sigma C)] \succeq 0, \quad (3.10)$$

and the independent nonzero Gram-gap fixture verifies that the difference cannot be dropped.

4. The physical response must precede pullback

At fixed conditional law the R-123 action owner is

$$\mathcal{P}_{\text{comp}} = -\frac{1}{2}\mathcal{T}, \quad q_{\text{comp}} = q_{\text{CN}} - \frac{1}{2}q_{\text{fut}}. \quad (4.1)$$

Combining R-128 and R-129 gives the exact bilinear identity

$$\boxed{q_{\text{comp}}(H, K) = \text{Re}\langle \mathbb{E}_f \dot{J}_H, \mathbb{E}_f \dot{J}_K \rangle + \text{Re}\langle \Phi, \mathbb{E}_f \ddot{J}_{H,K} \rangle - \frac{1}{2} D^2 \Theta[H, K].} \quad (4.2)$$

Thus the conditional-mean terms which canceled in q_{CN} return when the actual smaller action owner is reconstructed.

On a finite physical cylinder define the bilinear form

$$\beta_{\text{comp},h}(z, w) = \sum_{\text{owners}} \mathbb{E} q_{\text{comp},h}(Z_h)[z, w]. \quad (4.3)$$

For predictable synthesis L_{pred} , first take the physical Riesz response

$$A_{\text{comp},h}u = (\beta_{\text{comp},h}(\cdot, L_{\text{pred}}u))^{\sharp},$$

and only then pull back:

$$H_{\text{comp},h} = L_{\text{pred}}^* A_{\text{comp},h}. \quad (4.4)$$

In finite matrices, $A = q_{m\text{comp}}L$ and $H = L^* q_{m\text{comp}}L$. The two executable routes use different dimensions and rational matrices; both verify the action identity, selfadjoint pullback, vertical kernel, and orthogonal physical-shell forward/reverse sums.

Every physical pullback annihilates $\text{Ker } L$. In the primary fixture

$$L = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}, \quad v = (-1, 1, 1)^T \in \text{Ker } L,$$

one has

$$v^T H v = 0, \quad v^T \frac{9}{10} I v = \frac{27}{10}. \quad (4.5)$$

The source cost cannot be hidden inside the physical response. Moreover, (4.3)–(4.4) may initially define only a cylindrical or densely defined form. R-130 supplies no cutoff/chart-uniform bounded bare- L^2 extension.

5. Exact fixed-six-row conormal Gram envelopes

Let S_A be the three embedded real Pauli generators and put

$$\begin{aligned} q_A(u) &= \frac{u^T S_A u}{|u|^2 + e}, & T_A &= S_A - \alpha q_A I, \\ p_A &= 2S_A u, & \ell_A &= 2T_A u. \end{aligned} \quad (5.1)$$

Define

$$C_6(u)v = (\sqrt{c_0} p_A \cdot v, \sqrt{c_1} \ell_A \cdot v)_{A=1}^3, \quad B_e = C_6^T C_6. \quad (5.2)$$

The A1 coefficients diagonalize exactly:

$$a = c_0 + c_1(1 - \alpha)^2, \quad b = c_1\alpha(1 - \alpha), \quad c = c_1\alpha^2, \quad (5.3)$$

and hence

$$B_e(u) \preceq \beta_{\text{op}} |u|^2 I, \quad \beta_{\text{op}} = 4(a + 2b + c) = \frac{339}{2000P}. \quad (5.4)$$

Theorem 5.1 (pointwise Gram envelopes)

For every frozen real symmetric pointwise matrix Q ,

$$|D(B_e : Q)[a]| \leq L_6 \|Q\|_{\text{op}} |u| |a|, \quad (5.5)$$

$$\left| \frac{1}{2} D^2(B_e : Q)[a, a] \right| \leq H_6 \|Q\|_{\text{op}} |a|^2, \quad (5.6)$$

where

$$L_{\text{lin}} = 24c_0 = \frac{36}{125P}, \quad L_{\text{rat}} = 3c_1 \frac{3808}{81} = \frac{1071}{250P}, \quad \boxed{L_6 = \frac{1143}{250P}}, \quad (5.7)$$

and

$$H_{\text{lin}} = 12c_0 = \frac{18}{125P},$$

$$H_{\text{rat}} = 3c_1 \left(\frac{4624 + 5600 + 2240}{81} \right) = \frac{7011}{500P}, \quad \boxed{H_6 = \frac{7083}{500P}}. \quad (5.8)$$

For the rational row set

$$R_A = (S_A - q_A I)u, \quad \nabla q_A = \frac{2R_A}{d},$$

$$D^2 q_A = \frac{2(S_A - q_A I)}{d} - \frac{4(uR_A^T + R_A u^T)}{d^2}, \quad F_A = T_A - \alpha u (\nabla q_A)^T. \quad (5.9)$$

If $w_A = Q(T_A u)$ and $\beta_A = u^T Q(T_A u)$, the exact half-Hessian matrix of $\ell_A^T Q \ell_A$ is

$$\boxed{K_A(Q) = 4F_A^T Q F_A - 4\alpha \beta_A D^2 q_A - 4\alpha (\nabla q_A w_A^T + w_A \nabla q_A^T)}. \quad (5.10)$$

The elementary chain

$$|T_A u| \leq \frac{14}{9} |u|, \quad |u| |\nabla q_A| \leq 4, \quad \|F_A\| \leq \frac{34}{9}, \quad |u|^2 \|D^2 q_A\| \leq 20 \quad (5.11)$$

produces the three terms $4624/81$, $5600/81$, and $2240/81$. The primary route checks (5.10) symbolically on a two-real slice. The independent route derives the quotient with exact second-order jet arithmetic on a different three-real fixture and checks independent frozen rational oracles.

For the physical energy $E = (B_e : Q)/2$, the force, Hessian, and Taylor remainder constants are, respectively,

$$\frac{L_6}{2} = \frac{1143}{500P}, \quad H_6 = \frac{7083}{500P}, \quad \frac{H_6}{2} = \frac{7083}{1000P}. \quad (5.12)$$

This factor firewall is part of the theorem. If $Q = Q(u)$, the derivatives of Q must be added. Pointwise bounds (5.5)–(5.6) do not themselves imply a spatial multiplier, shell-decay, or uniform response theorem.

6. A floor-uniform separated rational H^2 no-go

Let

$$F_e(s) = \frac{s^3}{s^2 + e}, \quad G_e(x) = F_e(\sin x)$$

on the normalized circle. Its boundary profile

$$f(y) = \frac{y^3}{1 + y^2}$$

satisfies

$$f''(y) = \frac{2y(3 - y^2)}{(1 + y^2)^3}, \quad \int_{\mathbb{R}} |f''(y)|^2 dy = \frac{3\pi}{4}. \quad (6.1)$$

The two zero layers of $\sin x$ give

$$\lim_{e \downarrow 0} e^{1/2} \|G_e''\|_{L^2(\mathbb{T}, dx/(2\pi))}^2 = \frac{3}{4}, \quad (6.2)$$

and therefore

$$\|G_e''\|_2 \sim \frac{\sqrt{3}}{2} e^{-1/4}. \quad (6.3)$$

The primary route derives (6.1) exactly. The independent route uses an adaptive boundary-rescaled Simpson calculation down to $e = 10^{-6}$ and checks both coefficients.

Equations (6.2)–(6.3) rule out a floor-uniform estimate obtained by separating the rational coefficient and bounding D^2C in unweighted H^2 . They do not refute the complete Gram route: the cancellation between $C D^2C$ and $(DC)^2$ inside $D^2(C^T C)$ is load-bearing.

7. The sharp balanced bridge and its present boundary

For periodic vector-valued z , integration by parts gives

$$-\int |z|^2 z \cdot \Delta z = \int |z|^2 |\nabla z|^2 + 2 \sum_i \int (z \cdot \partial_i z)^2. \quad (7.1)$$

Cauchy–Schwarz proves (1.1). The constant is sharp: for

$$z = A(\cos Nx, \sin Nx) \quad (7.2)$$

both sides equal $A^4 N^2$. The independent route also checks (7.2) by a 4096-point vector-circle integration and verifies that a constant below one fails.

Let

$$A_0 = \frac{M_R^2}{c_{\text{sym}}}, \quad C_{\text{br}} = \sqrt{32A_0}. \quad (7.3)$$

The A1/A8 bridge converts the mixed term to

$$c_{\text{bal}} \int |z|^2 |\nabla z|^2 \leq c_{\text{bal}} C_{\text{br}} \sqrt{XY}, \quad (7.4)$$

up to the already recorded Gaussian baseline constant. It is payable by $\eta X + \zeta Y$ if

$$c_{\text{bal}} C_{\text{br}} \leq 2\sqrt{\eta\zeta}. \quad (7.5)$$

The reserve data are read from the R-103 executable authority, not copied as untracked literals:

$$\eta_{\text{av}} = \frac{197}{440} - \frac{3}{125P}, \quad \zeta_{\text{av}} = \frac{3}{25}. \quad (7.6)$$

The independently recomputed values are

$$C_{\text{br}} = 17.1842828361\dots, \quad c_{\text{bal,max}} = \frac{2\sqrt{\eta_{\text{av}}\zeta_{\text{av}}}}{C_{\text{br}}} = 0.0267957744\dots \quad (7.7)$$

The live full-cross Cartan factor is read from the R-124 executable authority. Its oriented half is

$$c_{\text{Cartan}}^{\text{or}} = \frac{1}{2} \frac{2680}{729} c_1 = \frac{67}{1200P} = 0.013958333\dots, \quad (7.8)$$

and

$$\frac{c_{\text{Cartan}}^{\text{or}}}{c_{\text{bal,max}}} = 0.5209154659\dots < 1. \quad (7.9)$$

In the full-cross convention both the coefficient and the acceptance denominator double, so the ratio is unchanged. This is only a local viability diagnostic. It is not a global upper bound on the complete production c_{bal} ; C_{mix} , C_{far} , R_0 , S_0 , and same-channel spending can change the available ceiling.

8. A centered-heat direct-low candidate

Let

$$\gamma_\ell(x) = \sum_i \text{Tr } \Gamma_{\ell,i}(x, x), \quad g_\ell = \|\gamma_\ell\|_\infty,$$

and let

$$s_\Sigma^2(x) = \mathbb{E}_r |r(x)|^2$$

for a deterministic or strict-past centered heat. For

$$Q_\ell(Z) = \frac{1}{2} \|\Phi_\ell(Z)\|^2 - \frac{1}{2} \Theta_\ell(Z), \quad (8.1)$$

discard only the positive mean square. The operator inequality (5.4), heat centering

$$\mathbb{E}_r |Z + r|^2 = |Z|^2 + s_\Sigma^2, \quad (8.2)$$

and the once-owned trace give

$$Q_\ell(Z) \geq -\frac{\beta_{\text{op}}}{2} g_\ell \|Z\|_2^2 - C_{\Sigma, \ell}, \quad (8.3)$$

where

$$C_{\Sigma, \ell} = \frac{\beta_{\text{op}}}{2} \int \gamma_\ell(x) s_\Sigma^2(x) \, dx. \quad (8.4)$$

For volume V and $Y = \|Z\|_6^6$,

$$\|Z\|_2^2 \leq V^{2/3} Y^{1/3}. \quad (8.5)$$

Set

$$a_\ell = \frac{\beta_{\text{op}}}{2} g_\ell V^{2/3}.$$

The sharp scalar maximization gives, for every $\zeta > 0$,

$$a_\ell Y^{1/3} \leq \zeta Y + \frac{2a_\ell^{3/2}}{3\sqrt{3}\zeta}, \quad (8.6)$$

and hence

$$\boxed{\mathbb{E} Q_\ell \geq -\zeta \mathbb{E} Y - \frac{2a_\ell^{3/2}}{3\sqrt{3}\zeta} - C_{\Sigma, \ell}.} \quad (8.7)$$

On the 16^3 torus,

$$\frac{a_\ell}{g_\ell} = \frac{2712}{125P} = 5.423999 \dots \quad (8.8)$$

The primary route checks the centered heat, trace operator factor, finite volume Holder equality case, and Young maximizer. The independent route checks the cubic volume and the global scalar maximum on a deterministic grid.

The conditional hypothesis in (8.2) fails for anticipative heat. More importantly, (8.7) is a new direct-low candidate after terminal recombination. It is not an identification with the R-079 conditional or complete-low owners and does not supply their D, K, R_0, S_0 data or the absolute baseline.

9. Complete-square Gram Schur and two exact failures

Let a complete positive square have the form

$$\|Wx + T\ell\|^2.$$

Its Gram blocks are

$$M_2 = W^*W, \quad K = W^*T, \quad D = T^*T. \quad (9.1)$$

When D is invertible,

$$\boxed{M_2 - KD^{-1}K^* = W^*(I - P_T)W \succeq 0.} \quad (9.2)$$

Thus the Douglas range relation for the internal low coupling is automatic. For noninvertible D , the same statement uses the reduced solution or closed-range formulation.

Semidefiniteness is not a strict gap. If $\text{Ran } W \subseteq \text{Ran } T$, the right side of (9.2) is zero. The primary rank-one fixture realizes zero exactly; the independent route uses different three-dimensional T, W , obtains a positive Schur fixture with determinant $75/2$, and then a distinct rank-one W_0 with zero Schur complement.

Nor is a childwise square refinement invariant:

$$1^2 + (-1)^2 = 2 > (1 - 1)^2 = 0. \quad (9.3)$$

The independent route repeats the obstruction with children 2, -2 and also checks the cross factor in the full block expansion. Children must be aggregated before the terminal quotient; a complete square alone does not give the strict augmented gap required by A13.

10. Proof-search evidence map

The reusable proof decisions are summarized below. Each substantive verdict also has an append-only exploration-ledger entry; the two broad no-go families are promoted to the negative registry.

Route	Verdict	Boundary or revisit condition
Post-recombined Ξ Doob recode	advanced	Fixed terminal pair, filtration, and trace partition only.
Legacy roots share the Ξ terminal	failed	Recombine first and rerun Doob.
Matching trace implies matching energy	failed	Revisit only with $\mu = 0$, $\Sigma_c = \Gamma$.
Average C before squaring heat	failed	Retain $P_\Sigma(C^T C)$.
Finite response before pullback	advanced	Uniform production extension open.
Hide $9I/10$ in physical response	failed	Source owner remains separate.
Pointwise six-row envelopes	advanced	Multiplier and shell estimates open.
Separated floor-uniform rational H^2	failed	Retain complete Gram cancellation.
Sharp balanced bridge	advanced	Complete global c_{bal} open.
Local Cartan budget	viable diagnostic	Compute all global owners before acceptance.
Centered-heat direct low	conditional advance	Prove owner equivalence and absolute anchor.
Complete Gram gives PSD Schur	advanced	Strict gap needs quantitative transversality.
Complete Gram gives strict gap	failed	Rank-one alignment has zero gap.
Childwise refinement squares	failed	Aggregate before quotient.

The main strategic consequence is narrow. The common-terminal coordinate, response order, pointwise Gram calculus, sharp balanced bridge, and direct-low candidate are now fixed. The next proof must estimate the complete owner response in the production spatial norms, calculate the global C_{mix} , C_{far} , and c_{bal} without double spending, and prove a transverse low gap or an equivalent absolute anchor. Repeating legacy rootwise compression, separated rational H^2 , or childwise square refinement cannot close those gates.

11. Devil’s-advocate review

1. **Objection: the Ξ identity transports every R-079 root block. UPHELD against that inference.** The compression is state-dependent; only a new post-recombined Doob decomposition is valid.
2. **Objection: matching trace gives a positive matching-energy packet. UPHELD.** Equation (3.8) retains a signed covariance defect unless $\mu = 0$ and $\Sigma_c = \Gamma$.
3. **Objection: the finite response fixture constructs the A1 production response. UPHELD.** It certifies the order of construction and kernel/source firewall only; the uniform production form estimate is open.
4. **Objection: pointwise Gram constants imply the shell bound. UPHELD.** Spatial multipliers, heat, chart derivatives, and shell decay still have to be controlled.
5. **Objection: the separated H^2 divergence refutes the complete Gram route. DISMISSED.** The no-go targets D^2C alone, while Theorem 5.1 keeps the cancellation in $D^2(C^TC)$.
6. **Objection: the local Cartan ratio closes c_{bal} . UPHELD.** It is one local owner and not the global coefficient.
7. **Objection: the direct-low bound closes R-079 low. UPHELD.** Owner identification, matching energy, D , K , R_0 , S_0 , and the absolute baseline remain missing.
8. **Objection: a complete Gram square gives a strict gap. UPHELD.** Rank-one alignment makes the Schur complement zero.
9. **Objection: the source cost belongs to the physical pullback. UPHELD.** The pullback vanishes on $\text{Ker } L$ while $9I/10$ does not.
10. **Objection: forward and reverse create two reserves. UPHELD.** They are one true-adjoint owner.
11. **Objection: R-130 closes A13 or Sector A. DISMISSED.** Both A13 gates, the production response, global balanced/low closure, and the subsequent Nelson chain remain open.

12. Executed evidence and reproduction

The primary route uses SymPy exact algebra and reads A1, A8, R-103, and R-124 upstream evidence directly. It derives all reported constants and writes an atomic JSON result. The independent route uses only the Python standard library, does not import or read the primary result, and uses different exact jet, eight-atom terminal, physical-response, heat-Gram, Schur, adaptive-floor, vector-circle, and global-Young fixtures.

Run

```
E:\Dev\TECT.venv\Scripts\python.exe \
  codes/foundations/a13_classii_terminal_xi_conormal_gram_balanced_low_response_boundary.py

E:\Dev\TECT.venv\Scripts\python.exe \
  codes/foundations/a13_classii_terminal_xi_conormal_gram_balanced_low_response_boundary_independent.py
```

The frozen child outputs are

```
R-130 primary PASS: 58/58
R-130 independent PASS: 90/90
```

The integrated verifier additionally checks authority and file hashes, semantic scopes, generated surfaces, exploration and negative records, source-to-PDF rebuilding, zero overfull boxes, extracted page identity, hash-pinned 130-DPI Poppler renders, recursive PDF security including associated and embedded files, all-page visual inspection, and deterministic JSON output.

13. Result footer

```
Result ID: A13-CLASSII-TERMINAL-XI-CONORMAL-GRAM-BALANCED-LOW-
RESPONSE-BOUNDARY (R-130).
Precise statement: fixed post-recombined Xi terminal and matching-trace
  recode in Section 3; finite-cylinder physical-response-before-pullback
  construction in Section 4; exact frozen-Q six-row Gram envelopes in
  Theorem 5.1; separated rational D2 floor no-go in Section 6; sharp
  constant-one balanced bridge in Section 7; centered-heat direct-low
  candidate in Section 8; complete-Gram PSD Schur identity and exact
  strict-gap/refinement failures in Section 9.
Scope: fixed finite cutoff and positive floor; fixed temporally faithful
  chart, conditional law, filtration, terminal pair, and once-owned
  covariance partition; finite matrices for the response fixtures; frozen
  pointwise Q for the six-row derivative envelopes.
Dependencies: A1, A8; R-079, R-082, R-103, R-104, R-120, R-121, R-123,
  R-124, R-128, and R-129.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
  codes/foundations/a13_classii_terminal_xi_conormal_gram_balanced_low_
  response_boundary_verify.py
Expected output: R-130 integrated PASS; primary 58/58 and independent 90/90;
  manifest, PDF, exploration, negative-result, and generated-surface gates
  pass with deterministic integrated JSON.
Falsification gate: failure of the Xi/Fierz recode or trace partition; heat
  Gram inequality; response action, adjoint, or kernel fixture; L6 or H6
  envelope; rational zero-layer asymptotic; balanced constant-one identity;
  centered-heat low bound; Schur identity or zero-gap/refinement fixtures;
  any authority/hash, PDF, exploration, registry, or public-surface check.
Tier before / after: T4 / T4; no promotion.
No-overclaim statement: no arbitrary legacy-root transport, unconditional
  matching energy, uniform production response, spatial multiplier/shell
  theorem, complete C_mix/C_far/c_bal, historical R-079 low closure, strict
  gap, absolute anchor, OVERLAP_src, Nelson, removal, interacting measure,
  Sector-A closure, or T5--T7 result follows.
Next required action: prove A13-CLASSII-OWNER-COMPLETE-TRACE-EXCESS-PHYSICAL-
  RESPONSE-FORWARD-BALANCED-LOW-BOUND, including the uniform complete response,
  C_mix, C_far, global c_bal, matching energy, low transversality or D/K/R0/S0,
  and the absolute anchor before invoking R-129 coanalysis.
Proof complete: false. Sector A closed: false.
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No-overclaim footer. R-130 proves a fixed post-recombination common-terminal coordinate, an abstract finite-cylinder response-before-pullback construction and kernel firewall, exact fixed-six-row pointwise Gram envelopes, a sharp deterministic balanced inequality, and a centered-heat direct-low candidate.

It does not prove a cutoff-, floor-, chart-, control-, filtration-, visit-, shell-, or refinement-uniform bounded production response; complete C_{mix} , C_{far} , or c_{bal} ; historical R-079 low-owner closure; a strict augmented gap; the production common-terminal matching-energy theorem or absolute anchor; $\text{OVERLAP}_{\text{src}}$; Nelson; removals; an interacting measure; Sector-A closure; or T5–T7 promotion.